

1. [20 pts] Please complete the code of LMS algorithm. Hint:
 - h : estimated impulse response
 - x : input signal (far-end reference)
 - d : desired signal (far-end recorded + near-end recorded)
 - e : error signal (estimated near-end)
2. [20 pts] (Continuing from 1). Please choose an appropriate filter length and step-size. Describe in detail how you choose these parameters.
3. [20 pts] Plot the impulse response $h(t)$ and the learning curve. Describe what you observe in these plots.
4. [20 pts] Is there an upper limit for the step-size α ? Or can you come up with another way of choosing α ?
5. [20 pts] Discuss what would happen if you change filter length or step-size.

For this homework, please turn in your code and a brief report.